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Tutorial Course for Intro to Sensor-Integrated Systems

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Basic DC Bridge Analysis and Calibration

Core Concept: Why do we need a bridge?

- **Pain Point:** The output of sensors (such as RTDs) is typically a **minute change in resistance**, which instruments cannot read directly.
- **Function:** A Deflection Bridge is a conversion circuit responsible for accurately transforming "changes in resistance" into "measurable voltage differences."

Key Parameters and Symbol Definitions

- V_s (**Supply Voltage**): The reference voltage driving the system, determining the upper output limit and power consumption.
- E_{Th} (**Output Voltage**): The Thévenin equivalent potential difference read at the measurement terminals.
- R_1 or R_T : **Active Bridge Arm** (i.e., the sensor itself, whose resistance changes with temperature, etc.).
- R_2, R_3, R_4 : **Fixed Bridge Arms** (precision reference resistors selected by the engineer).

Three Core Formulas

- **Output Voltage Formula:**

$$E_{Th} = V_s \left(\frac{R_1}{R_1 + R_4} - \frac{R_2}{R_2 + R_3} \right)$$

- **Bridge Balance Condition:**

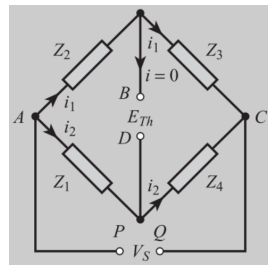
When the required output $E_{Th} = 0$, the cross-ratio equality must be met:

$$\frac{R_4}{R_1} = \frac{R_3}{R_2}$$

- **Temperature Characteristics of Metal Resistance (RTD (Resistance Temperature Detector) Sensor Formula):**

$$R_T = R_0(1 + \alpha T)$$

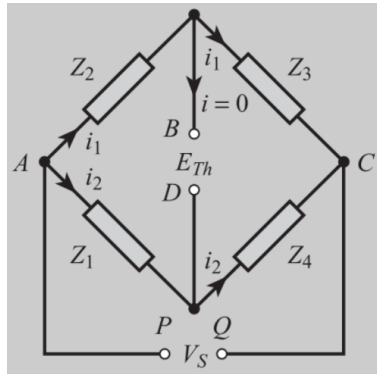
(Note: R_0 is the reference resistance at 0°C , and α is the temperature coefficient)



Exercise

A deflection bridge has a supply voltage of 10 V. Find the bridge output voltage when:

- (a) $R_1 = 101\ \Omega$, $R_2 = 99\ \Omega$, $R_3 = 101\ \Omega$, $R_4 = 99\ \Omega$
- (b) $R_1 = 100.1\ \Omega$, $R_2 = 99.9\ \Omega$, $R_3 = 100.1\ \Omega$, $R_4 = 99.9\ \Omega$.



Solution

(a)

$$E_{Th} = 10 \times \left(\frac{101}{101 + 99} - \frac{99}{99 + 101} \right)$$

$$E_{Th} = 10 \times \left(\frac{101}{200} - \frac{99}{200} \right)$$

$$E_{Th} = 10 \times \left(\frac{2}{200} \right) = 10 \times 0.01 = 0.1 \text{ V}$$

(b)

$$E_{Th} = 10 \times \left(\frac{100.1}{100.1 + 99.9} - \frac{99.9}{99.9 + 100.1} \right)$$

$$E_{Th} = 10 \times \left(\frac{100.1}{200} - \frac{99.9}{200} \right)$$

$$E_{Th} = 10 \times \left(\frac{0.2}{200} \right) = 10 \times 0.001 = 0.01 \text{ V}$$

Exercise

A platinum resistance sensor has a resistance of $100\ \Omega$ at $0\ ^\circ\text{C}$ and a temperature coefficient of resistance of $4 \times 10^{-3}\ ^\circ\text{C}^{-1}$.

- (a) The above sensor is incorporated into a bridge circuit which has $R_3/R_2 = 100$. Find the value of R_4 such that $V_{\text{OUT}} = 0$ at $0\ ^\circ\text{C}$.
- (b) Complete the bridge design by calculating the supply voltage required to give $V_{\text{OUT}} = 100\ \text{mV}$ at $100\ ^\circ\text{C}$.

Solution

Known Parameters:

- Initial resistance of the Platinum resistance sensor at 0°C : $R_0 = 100\ \Omega$
- Temperature Coefficient of Resistance (TCR): $\alpha = 4 \times 10^{-3}\ ^{\circ}\text{C}^{-1}$

(a) Find the value of R_4 such that $V_{OUT} = 0$ at 0°C .

Solution:

At 0°C , the resistance of the sensor is $R_T = R_0 = 100\ \Omega$.

The problem requires the output voltage $V_{OUT} = 0$ at this time, which means the bridge is in a **balanced state**.

According to the balance condition of a deflection bridge (Textbook Formula [9.9]):

$$\frac{R_4}{R_T} = \frac{R_3}{R_2}$$

Given the bridge arm ratio $\frac{R_3}{R_2} = 100$, substitute the known data into the formula:

$$\frac{R_4}{100} = 100$$

$$R_4 = 100 \times 100 = 10,000\ \Omega$$

Solution

(b) Complete the bridge design: Calculate the required supply voltage V_s to output 100 mV at 100°C.

Solution:

Step 1: Calculate the sensor resistance R_{100} at 100°C.

Based on the linear resistance-temperature formula:

$$R_T = R_0(1 + \alpha T)$$

$$R_{100} = 100 \times (1 + 0.4) = 140 \, \Omega$$

Step 2: Use the exact output voltage formula of the bridge to solve for V_s .

The general output voltage formula for a deflection bridge (Textbook Formula [9.6]) is rearranged as:

$$V_{OUT} = V_s \left(\frac{R_T}{R_T + R_4} - \frac{1}{1 + R_3/R_2} \right)$$

Given the target output voltage $V_{OUT} = 100 \, \text{mV} = 0.1 \, \text{V}$, substitute all known parameters:

$$0.1 = V_s \left(\frac{140}{140 + 10,000} - \frac{1}{1 + 100} \right)$$

$$V_s = \frac{0.1 \times 51,207}{200} = \frac{5,120.7}{200} \approx 25.6 \, \text{V}$$

Engineering Design and Linearization of Sensor Bridges

1. Core Pain Point: The Gap Between Ideal and Reality

- **Ideal State:** We hope for a perfectly proportional, linear relationship between the input (e.g., temperature) and the output voltage.
- **Reality:** The **voltage division characteristics** of the bridge itself possess inherent nonlinearity; furthermore, some sensors (such as thermistors) exhibit **severe exponential nonlinearity**.
- **Engineering Goal:** Without relying on software algorithms, use the clever configuration of **hardware circuit parameters** to "force" the output curve into a straight line.

2. Strategy One: Large Bridge Arm Ratio Method (For minute changes like RTD Platinum resistors)

- **Design Concept:** Make the fixed bridge arm resistance much larger than the sensor resistance (i.e., bridge ratio $r = R_3/R_2 \gg 1$). Use a massive fixed base value to "dilute/mask" the fluctuations in the denominator.
- **Engineering Trade-off:** Sacrifice system sensitivity (resulting in a smaller output voltage) in exchange for **excellent linearity**.
- **Engineering Approximation Formula:**

$$V_{OUT} \approx V_s \frac{1}{r} \alpha T$$

3. Strategy Two: Three-Point Design Method (For severe nonlinearity like Thermistors)

- **Design Concept:** Utilize the "upward curvature" of the bridge output curve to precisely cancel out the "downward curvature" caused by the exponential drop in thermistor resistance ($R_\theta \propto e^{1/T}$).
- **Three-Step Practical Guide:**
 1. Lock in the **Start Point, Midpoint, and End Point** of the measurement temperature range.
 2. Mandate that these three points output perfectly **equidistant voltages** (e.g., 0 V, 0.5 V, 1.0 V).
 3. Set up a system of equations for the three states to solve for the unique **Bridge Ratio r** (typically the optimal r value for thermistors is between 0.25 ~ 0.30) and the **Supply Voltage V_s** .

Exercise

A platinum resistance sensor has a resistance of $100\ \Omega$ at $0\ ^\circ\text{C}$ and a temperature coefficient of resistance of $4 \times 10^{-3}\ ^\circ\text{C}^{-1}$. Given that a $15\ \text{V}$ supply is available, design a deflection bridge giving an output range of 0 to $100\ \text{mV}$ for an input range of 0 to $100\ ^\circ\text{C}$:

- (a) using the procedure summarised by eqns [9.7] and [9.8];
- (b) using the linear approximation of eqn [9.15].

Give values for all circuit components and assume a high impedance load.

- (c) How should the circuit be altered if the input range is changed to 50 to $150\ ^\circ\text{C}$?

$$V_{\text{MIN}} = V_S \left(\frac{1}{1 + R_4/R_{I\text{MIN}}} - \frac{1}{1 + R_3/R_2} \right) \quad [9.7]$$

$$V_{\text{MAX}} = V_S \left(\frac{1}{1 + R_4/R_{I\text{MAX}}} - \frac{1}{1 + R_3/R_2} \right) \quad [9.8]$$

$$E_{Th} = V_S \frac{R_2}{R_3} \alpha T \quad [9.15]$$

Solution

Known Initial Parameters:

- Platinum resistor value at 0°C: $R_0 = 100 \, \Omega$
- Temperature Coefficient of Resistance (TCR): $\alpha = 4 \times 10^{-3} \, ^\circ\text{C}^{-1}$
- Supply Voltage: $V_s = 15 \, \text{V}$
- Input Temperature Range: 0°C to 100°C
- Target Output Voltage Range: 0 V to 100 mV (0.1V)

First, calculate the sensor resistance values at the endpoints of the temperature range using

$$R_T = R_0(1 + \alpha T):$$

- At $T_{MIN} = 0^\circ\text{C}$: $R_{MIN} = 100 \, \Omega$
- At $T_{MAX} = 100^\circ\text{C}$: $R_{MAX} = 100(1 + 0.004 \times 100) = 140 \, \Omega$

(a) Exact Design using Formulas [9.7] and [9.8]

Let the bridge arm ratio be $r = \frac{R_3}{R_2}$.

1. Based on Zero-Point Balance Condition (Output is 0V at 0°C):

According to formula [9.7], the simplified balance equation [9.9] is:

$$\frac{R_4}{R_{MIN}} = \frac{R_3}{R_2} = r \implies R_4 = r \cdot R_{MIN} = 100r$$

2. Based on Full-Scale Condition (Output is 0.1V at 100°C):

Substitute into the general deflection bridge formula [9.8]:

$$0.1 = 15 \left(\frac{1}{1 + R_4/R_{MAX}} - \frac{1}{1 + r} \right)$$

Cross-multiply to obtain a quadratic equation:

$$r^2 + 2.4r + 1.4 = 60r \implies r^2 - 57.6r + 1.4 = 0$$

$$r = \frac{57.6 + \sqrt{57.6^2 - 4 \times 1.4}}{2} \approx 57.58$$

3. Calculate Specific Component Values:

- $R_4 = 100 \times 57.58 = 5758 \, \Omega$
- Assuming $R_2 = 100 \, \Omega$
 $R_3 = 5758 \, \Omega$.

Solution

(b) Design using the Linear Approximation Method from Formula [9.15]

The linear approximation formula [9.15] in the textbook is built on the assumption that $r \gg 1$:

$$E_{Th} \approx V_s \frac{R_2}{R_3} \alpha T = V_s \frac{1}{r} \alpha T$$

1. Determine r based on Full-Scale Requirements:

Substitute $T = 100^\circ\text{C}$ and $E_{Th} = 0.1\text{ V}$:

$$0.1 = 15 \times \frac{1}{r} \times (4 \times 10^{-3}) \times 100$$

$$0.1 = \frac{15 \times 0.4}{r} = \frac{6}{r}$$

$$r = 60$$

Solution

(c) How should the circuit change if the input range is changed to 50°C to 150°C?

We need to redefine the resistance range:

- New start point $T_{MIN} = 50^{\circ}\text{C} \implies R_{MIN} = 100(1 + 0.004 \times 50) = 120 \Omega$
- New end point $T_{MAX} = 150^{\circ}\text{C} \implies R_{MAX} = 100(1 + 0.004 \times 150) = 160 \Omega$

The circuit requires two key modifications:

1. **Change the Zero-Balance Resistor (R_4):**

To output 0V at 50°C (when sensor resistance is 120 Ω), the balance condition changes to $R_4 = r \cdot R_{MIN} = 120r$.

2. **Adjust the Bridge Arm Sensitivity Ratio (r):**

Although the temperature span is still 100 degrees (and the resistance span is still 40 Ω), the **relative change rate** has changed. Therefore, r must be recalculated to ensure the full scale remains 100 mV.

- **Using Linear Approximation:** Full-scale formula becomes $E_{Th} \approx V_s \frac{1}{r} \left(\frac{\Delta R}{R_{MIN}} \right)$

$$0.1 = 15 \times \frac{1}{r} \times \frac{40}{120} \implies 0.1 = \frac{5}{r} \implies \mathbf{r = 50}$$

$$\text{Corresponding } R_4 = 50 \times 120 = \mathbf{6000 \Omega}.$$

- **Using Exact Method:** $0.1 = 15 \left(\frac{160}{160+120r} - \frac{1}{1+r} \right) \implies 3r^2 - 143r + 4 =$

$$0 \implies \mathbf{r \approx 47.64}$$

$$\text{Corresponding } R_4 = 47.64 \times 120 = \mathbf{5717 \Omega}.$$

Exercise

The resistance R_θ Ω of a thermistor varies with temperature θ K according to the following equation:

$$R_\theta = 0.0585 \exp\left(\frac{3260}{\theta}\right)$$

Design a deflection bridge, incorporating the thermistor, to the following specification:

- Input range 0 to 50 °C
- Output range 0 to 1.0 V
- Relationship between output and input to be approximately linear.

Solution

Step 1: Calculate thermistor resistance at key temperature points

The known thermistor formula is: $R_{\theta} = 0.0585 \exp\left(\frac{3260}{\theta}\right)$.

Convert Celsius to Kelvin ($\theta = T + 273$) and calculate the resistance at 0°C, 25°C (midpoint), and 50°C:

- **Start Point (0°C = 273 K):**

$$R_{273} = 0.0585 \exp\left(\frac{3260}{273}\right) = 0.0585 \exp(11.9414) \approx 8979.2 \, \Omega$$

- **Midpoint (25°C = 298 K):**

$$R_{298} = 0.0585 \exp\left(\frac{3260}{298}\right) = 0.0585 \exp(10.9396) \approx 3297.3 \, \Omega$$

- **Endpoint (50°C = 323 K):**

$$R_{323} = 0.0585 \exp\left(\frac{3260}{323}\right) = 0.0585 \exp(10.0929) \approx 1413.9 \, \Omega$$

Solution

Step 2: Establish the system of equations for linearization conditions

The output voltage formula for a deflection bridge is:

$$E_{Th} = V_s \left(\frac{1}{1 + R_4/R_\theta} - \frac{1}{1 + R_3/R_2} \right)$$

Let the bridge arm fixed resistance ratio be $r = R_3/R_2$. We mandate that the output voltage satisfies an absolute linear relationship:

1. **At 0°C (273 K), output is 0 V (Zeroing condition):**

$$0 = V_s \left(\frac{1}{1 + R_4/R_{273}} - \frac{1}{1 + r} \right) \implies \frac{R_4}{R_{273}} = r \implies R_4 = r \cdot R_{273}$$

2. **At 25°C (298 K), output is 0.5 V (Forced linearity):**

$$0.5 = V_s \left(\frac{1}{1 + r \frac{R_{273}}{R_{298}}} - \frac{1}{1 + r} \right)$$

3. **At 50°C (323 K), output is 1.0 V (Full-scale condition):**

$$1.0 = V_s \left(\frac{1}{1 + r \frac{R_{273}}{R_{323}}} - \frac{1}{1 + r} \right)$$

Solution

Step 3: Solve for the bridge ratio r

To simplify the calculation, first calculate the resistance ratio coefficients:

- $a = \frac{R_{273}}{R_{298}} = \frac{8979.2}{3297.3} \approx 2.7232$
- $b = \frac{R_{273}}{R_{323}} = \frac{8979.2}{1413.9} \approx 6.3507$

Divide the 1.0V equation by the 0.5V equation to eliminate V_s :

$$\frac{1.0}{0.5} = 2 = \frac{\frac{1}{1+br} - \frac{1}{1+r}}{\frac{1}{1+ar} - \frac{1}{1+r}} \quad 4.2456r = 1.1051 \implies r \approx \mathbf{0.2603}$$

Step 4: Calculate remaining component parameters

1. Calculate Zeroing Resistor R_4 :

$$R_4 = r \cdot R_{273} = 0.2603 \times 8979.2 \, \Omega \approx \mathbf{2337 \, \Omega}$$

2. Calculate Supply Voltage V_s :

Substitute $r = 0.2603$ and coefficient b into the 1.0 V equation:

$$1.0 = V_s \left(\frac{1}{1 + 6.3507 \times 0.2603} - \frac{1}{1 + 0.2603} \right)$$
$$1.0 = -0.4166 \cdot V_s \implies \mathbf{V_s \approx -2.40 \, V}$$

3. Determine R_2 and R_3 :

The ratio is fixed at $R_3/R_2 = 0.2603$. To limit current and power consumption, reference resistors of a reasonable order of magnitude can be freely chosen. For example:

- $R_2 = 10 \, \text{k}\Omega$ (10,000 Ω)
- $R_3 = 2.6 \, \text{k}\Omega$ (2603 Ω)

System-Level Integration and Signal Amplification

1. Core Pain Point: Why is a bridge alone not enough?

- **Reality Bottleneck:** The bridge output (E_{Th}) derived in the previous sections is typically extremely weak, often only a few millivolts (mV) or even microvolts (μV).
- **Industrial Requirement:** Backend microcontrollers, ADCs, or PLCs usually require standard voltage signals of 1.0 V to 5.0 V. The gap between the two is by a factor of hundreds!
- **The Solution:** An **Amplifier** must be cascaded at the backend of the bridge.

2. Core Weapon: Ideal Operational Amplifier (Op-Amp)

The cornerstone of modern signal conditioning. Always remember its two "magic tricks" (ideal assumptions) when solving problems:

- **"Virtual Open" ($i_+ = i_- = 0$):** The input impedance of the op-amp is infinite. This means it will **never "steal" any current** from the bridge, perfectly ensuring that the bridge formulas we learned earlier remain valid!
- **"Virtual Short" ($V_+ = V_-$):** Under negative feedback, the potentials at the two input terminals of the op-amp are always forced to be equal.

3. System Integration: Bridge + Amplifier = Ultimate Formula

When a Four-Arm Active Bridge is connected to a **Differential Amplifier**, the final integrated output formula is:

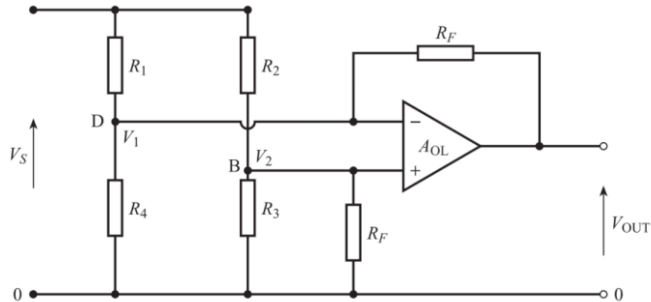
$$V_{OUT} = 2 \frac{R_F}{R_0} V_S G e$$

- V_{OUT} : Final standard output voltage (e.g., 1.0 V)
- R_F : Feedback resistor of the op-amp (**The only component engineers need to calculate and select!**)
- R_0 : Initial reference resistance of the sensor (e.g., 120 Ω)
- V_S : Bridge supply voltage (e.g., 15 V)
- G : Sensitivity coefficient of the sensor (Gauge Factor)
- e : Minute physical variable (e.g., mechanical Strain)

Exercise

A strain in the range 0 to 10^{-3} is to be measured using a four-strain-gauge bridge connected to a differential amplifier (Figure 9.11). The strain gauges have resistance $120\ \Omega$ and gauge factor 2.1, and the supply voltage is 15 V. Find the value of the amplifier feedback resistance if the output signal range is 0 to 1.0 V.

Figure 9.11 Strain gauge bridge connected to differential amplifier.



Solution

Known Parameters:

- Upper limit of strain measurement range: $e = 10^{-3}$ (i.e., $1000 \mu\epsilon$)
- Resistance of strain gauge with no strain: $R_0 = 120 \Omega$
- Sensitivity coefficient (Gauge factor): $G = 2.1$
- Bridge supply voltage: $V_S = 15 \text{ V}$
- Target full-scale output voltage: $V_{OUT} = 1.0 \text{ V}$

Substitute all known values into the formula:

$$R_F = \frac{1.0 \times 120}{2 \times 15 \times 2.1 \times 10^{-3}}$$

$$R_F \approx 1904.76 \Omega$$

1. **Bridge Output:** In a full-bridge configuration (one pair of opposite arms in tension, the other pair in compression), the bridge's own differential output voltage ($V_2 - V_1$) is $V_S G e$.
2. **Bridge Equivalent Internal Resistance:** Looking in from the output terminals, the Thévenin equivalent resistance of the bridge is $R_{IN} \approx R_0/2$.
3. **Differential Amplifier Gain:** The closed-loop gain is R_F/R_{IN} .

$$V_{OUT} = 2 \frac{R_F}{R_0} V_S G e$$

The problem requires us to find the feedback resistor R_F for a given strain e and a specific output voltage V_{OUT} . Rearranging the formula:

$$R_F = \frac{V_{OUT} \cdot R_0}{2 \cdot V_S \cdot G \cdot e}$$